

OSAMA HATEM

Aspiring Mechatronics & AI Engineer | Independent Researcher | Systems Builder

Kafr El-Sheikh, Egypt • Arabic (Native) • English (C1/C2 Fluent) • Portfolio / GitHub / LinkedIn

PROFILE

Self-directed STEM student building at the intersection of engineering, artificial intelligence, data, and physical systems. I learn by turning questions into working systems: from researching modular triboelectric energy harvesting, to building an AI-powered fire detection system, to developing a tested research-data pipeline that collects and organizes academic papers. Strongest in Python, systems thinking, independent research, and translating theory into practical prototypes.

EDUCATION

STEM High School – Kafr El-Sheikh | 2024–2027 (Expected)

Rigorous boarding-school program focused on mathematics, physics, engineering, and scientific problem solving. Relocated from home to pursue a more challenging academic environment.

RESEARCH & ENGINEERING PROJECTS

Energy Blocks | *Independent Research Project • Solo Researcher* | 2025–Present

Modular low-frequency triboelectric nanogenerator (TENG) system designed to convert wasted mechanical motion into usable energy for low-power devices.

- Developed a charge-transfer-first physical model focused on charge-per-cycle and energy-per-cycle rather than inflated voltage metrics.
- Designed modular FEP/nylon TENG units with copper–graphene composite electrodes and a complete rectification, storage, regulation, and output architecture.
- Validated 5.1 μC charge per cycle, 44.1 μA per cycle, stable capacitor charging, and linear current scaling across modules.
- Simulated electrical behavior in Python/Matplotlib, designed enclosures in Blender, and validated circuits in CircuitLab.
- Completed a research paper and academic poster; advanced to the final stage of the local ISEF selection competition.

AI Fire Detector | *Computer Vision / IoT / Cloud System* | 2025–Present

Real-time fire detection system using computer vision and IoT monitoring, designed to provide live feedback while minimizing cloud-storage and processing costs.

- Designed a cloud-based architecture for continuous video analysis using YOLOv8n with event-driven clip storage rather than storing 24/7 footage.
- Worked through system-level tradeoffs involving inference frequency, GPU availability, storage lifecycle policies, network transfer, and per-user operating cost.
- Integrated IoT device requirements with cloud processing and real-time alerting concepts.

Research Data Pipeline | *Python / Django / SQLite / OpenAlex API* | 2026

Tested research-data collection pipeline that retrieves academic papers from OpenAlex, normalizes metadata, stores papers and authors in SQLite, and exposes the workflow through both a CLI and a Django web interface.

- Built paginated collection with separate tracking for fetched, new, existing, skipped, and failed papers.
- Implemented retry and rate-limit handling for API failures and designed the collector to continue searching beyond already-stored results.
- Added automated tests for normalization, API behavior, database repositories, collection processing, and CLI functionality.

SELECTED PROJECTS & ACTIVITIES

Personal Productivity System | *Notion Second Brain* | 2024–Present

Designed a structured knowledge-management system for task management, goal tracking, learning logs, and knowledge capture.

School Club | *Founder & Leader* | 2024–Present

Founded and led a school club, developing skills in organization, delegation, communication, and community building.

Programming & Software Projects | *Independent Development* | 2024–Present

Built projects in Python and web technologies as a way to turn personal problems and ideas into working systems, progressing from small productivity tools to data pipelines, AI systems, and engineering research.

TECHNICAL SKILLS

Programming & Software: Python (Advanced), Data Structures & Algorithms, Git & Version Control, HTML, CSS, JavaScript, Django

Data & Analysis: Pandas, NumPy, Matplotlib, SQL, SQLite, Excel, data cleaning, API data collection, statistical analysis

AI & Computer Vision: YOLOv8, computer vision, machine learning fundamentals, neural networks, AI systems

Systems & Cloud: REST APIs, IoT architectures, Docker, networking fundamentals, cloud architecture, observability concepts

Engineering & Research: Python modelling, triboelectric systems, power-conditioning design, CircuitLab, Blender 3D modelling, academic writing

Mathematics: Calculus I & II, Linear Algebra, Probability & Statistics

Tools: Tkinter, Notion (Advanced), GitHub, pytest

ACHIEVEMENTS, PROGRAMS & ACTIVITIES

- ISEF Local Competition — Advanced to Final Stage with Energy Blocks (Solo Researcher)
- IPHyC — Finalist; top 15th percentile globally
- IEEE IC-ESL — National Finalist
- STEM School Excellence Recognition
- YRI Participant
- DECI — Digital Egypt Cubs Initiative, Levels 2–4
- Hackathon Participant and Independent Business Plan Author
- English School Club — Founder
- Lumiere Research Scholar Program — Applicant (Financial Aid Track)
- Pioneer Research Institute — Applicant
- Blue Belt — Kung Fu Martial Arts

CERTIFICATIONS & LEARNING

- Data Analysis Certification
- Fundamentals of Digital Marketing
- Prompt Engineering Certification
- Khan Academy — Calculus I & II, Linear Algebra, Probability & Statistics
- Self-directed study — Advanced Python, Data Structures & Algorithms, Git, HTML/CSS/JavaScript
- AI & Machine Learning Courses — Ongoing

LANGUAGES & INTERESTS

Arabic — Native | English — C1/C2 Fluent | Interests: AI, energy systems, physics, research, piano, digital art, reading, and creative system design.

LINKS

Portfolio: [insert final portfolio URL] • GitHub: [insert GitHub URL] • LinkedIn: [insert LinkedIn URL] • Project documentation: [insert documentation URL]